WHITE PAPER · HWACHEON MEGA 100 × INDUSTRIAL & GENERAL MANUFACTURING

Hwacheon Mega 100 for Industrial & General Manufacturing

Heavy-duty CNC lathe with industry-leading 10" thru-hole for tubing, collars, subs, and long-shaft work. This paper covers what the machine does, the industrial & general manufacturing parts we produce with it, the alloys we run, the tolerances and finishes industrial & general manufacturing buyers expect, and the documentation package.

16" OD × 100" L × 10" thru-hole
ENVELOPE

±0.0005" routine on critical features
TOLERANCE

1994
SINCE

Same-week / rig-down direct
RESPONSE



Texaco Thunder · 16" OD × 100" L × 10" thru-hole · heavy-duty CNC lathe with 10" thru-hole

PUBLISHED
January 15, 2026

LAST UPDATED
2026-01-15

FORMAT
Web + PDF

READ TIME

~6 min

Executive summary

Machine: Hwacheon Mega 100 (shop nickname: *Texaco Thunder*). heavy-duty CNC lathe with 10" thru-hole at B&R Productions, running from our New Waverly, Texas shop.

Application: Heavy-duty CNC lathe with industry-leading 10" thru-hole for tubing, collars, subs, and long-shaft work for Industrial & General Manufacturing customers — Industrial equipment OEMs, custom machinery builders, pump and valve manufacturers outside oilfield, food and pharma equipment suppliers, mining and heavy-equipment operators, reverse-engineering customers, sustainment for out-of-production industrial equipment.

Envelope: 16" OD × 100" L × 10" thru-hole. **Tolerance:** ±0.0005" routine on critical features. **Traceability:** Material by heat and lot on every job.

The machine, in context

The 10-inch through-hole is where Texaco Thunder earns its keep. Long tubing stock, downhole tool bodies with internal passages, coiled-tubing tool bodies, large valve stems with bore access from both ends — parts that other shops outsource because their machines can't accommodate the stock. This machine turns outsourced-work into in-house work.

Why Texaco Thunder for Industrial work

Industrial tubular work — long tubes, sleeves, cylinder bodies, through-bored housings — hits the same spindle-bore wall as oilfield parts do. A 10" through-hole at 16" OD × 100" L means the stock is fed through and worked in one piece rather than cut, chucked short, and flipped.

What this looks like on a real part

A long cylinder body or sleeve that needs machining at both ends and a consistent bore is the archetype. Cut it and flip it and the two ends are now related by however well it was re-indicated; feed it through and they're related by the original setup. For replacement parts going back into existing equipment, that concentricity is usually the requirement that determines whether the part actually fits or merely measures correctly.

> **When this is the wrong machine**
>
> 400/750 RPM. Small or speed-sensitive industrial turning belongs on Bearkat Blur, and the 4 or 8-station turret means complex multi-tool parts sit better elsewhere.

Full specifications

Model	Hwacheon Mega 100 ("Texaco Thunder")
Type	Heavy-duty flatbed horizontal CNC lathe — Mega-100 platform
Max turning capacity	16" OD × 100" L (B&R working spec)
Platform max cutting Ø	27.5" (700 mm) on the Mega-100
Chuck swing over bed	42.5" (1,080 mm)
Distance between centers	10 ft (3,000 mm) B&R config; platform available to 20 ft (6,000 mm)
Spindle bore	10" (255 mm) B&R config; platform range 6" – 12.6"
Spindle speed	400 / 750 RPM (two-range)
Spindle motor	25 HP
Turret	4 or 8-station heavy-duty
Machine weight	~35,000 lb (16,000 kg)
Purpose	Long shafts, main-shaft work, large pipe / tubing components
Tolerance capability	±0.0005" routine on critical features
Materials	Full oilfield alloy range — Inconel, Duplex 2205/2507, 17-4 PH, 4140, Monel

Values marked "platform" reflect the machine manufacturer's published spec range for this platform family; B&R's working spec is what we quote on this specific unit.

Who this serves in Industrial & General Manufacturing

Typical buyer: Industrial equipment OEMs, custom machinery builders, pump and valve manufacturers outside oilfield, food and pharma equipment suppliers, mining and heavy-equipment operators, reverse-engineering customers, sustainment for out-of-production industrial equipment.

What's hard about industrial & general manufacturing work: Legacy parts with no OEM support, one-off replacement components, small-lot production runs, materials that don't fit standard shop capabilities, and tight lead times on repair work where a down-machine is losing money by the day.

Typical Industrial & General Manufacturing parts we produce on Texaco Thunder

- Long hydraulic cylinders for industrial equipment
- Long spindle and shaft bores
- Rolling mill and press component bores
- Heat exchanger tube sheets and headers
- Custom deep-bored replacement parts

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

Alloys we run for Industrial & General Manufacturing

Standard range: Stainless (304, 316, 17-4 PH) · Tool steels · 4140 / 4340 · Brass · Bronze · Aluminum · Cast iron · Specialty alloys per customer requirement.

Industrial work covers the widest alloy range because the applications are the widest. Food and pharma often specify 316L or 17-4 PH; mining and heavy equipment specify tool steels and 4140/4340; pump manufacturers outside oilfield use everything from bronze bushings to stainless impellers. We work with the customer's spec sheet — if you tell us the alloy, we'll tell you honestly whether we can machine it and how quickly.

> **What "we run these weekly" means:** we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific to each alloy family. Your part is not the one we're experimenting on.

How we machine it — our 5-step process

This is the process B&R runs on every job, industrial & general manufacturing or otherwise. The specifics of feeds, speeds, and tooling shift by alloy and machine; the discipline is universal.

1. **Material verification.** Every job starts with material traceability. Heat and lot numbers documented against the mill test report. For customer-supplied material, we verify against the CofC before touching the stock.

2. **First-article layout.** Stock is measured and laid out before any cutting starts. Concentricity, roundness, and squareness of the starting material established; if the stock is out of tolerance we call the customer before wasting time on a bad blank.

3. **Roughing with process control for the alloy.** Feeds, speeds, and coolant strategy set for the specific alloy family — sharp coated carbide (or PCD for high-volume Ti Grade 5), positive rake, high-pressure through-tool coolant on 2507 and Inconel to control cutting temperature. Interrupted cuts get extra attention because they cycle heat into the workpiece.

4. **Finish pass to spec.** Finish pass with a fresh insert, controlled DOC, and measured surface-finish targets. Critical features (bore concentricity, seat grooves, sealing surfaces) are held tighter than most job-shop lathes can offer.

5. **CMM verification + documentation.** Every critical dimension CMM-verified. First-article report generated. Certificate of conformance issued with delivery. If your quality group needs a documentation package beyond that, tell us with the RFQ and we'll be straight about what we can and can't produce in-house.

Tolerance and finish expectations in Industrial & General Manufacturing

±0.0005" routine on critical features. Industrial applications range from precision (pump seal areas, close-fit bushings) to structural (mining and heavy-equipment brackets); we adjust the inspection plan to what the part actually needs. Surface finish measured on parts where finish drives function.

Documentation and quality workflow

Documentation calibrated to customer requirement. Material traceability standard for critical parts. Certificates of conformance issued with delivery. NDAs on request for proprietary designs or reverse-engineered legacy parts.

Response time and emergency work

Down-machine and production-critical work prioritized. Lead times honestly quoted — no false promises. For legacy reverse-engineering, we typically confirm feasibility on the first phone call. Call (936) 291-7827.

How we compare

Compared to sourcing a new OEM part: we're often faster and cheaper for one-offs and small runs, and for legacy parts where the OEM has moved on we're often the only realistic option. Compared to a cheapest-bidder job shop: we hold tolerance, document material, and don't ruin the part.

Working with B&R Productions

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request, and prints stay in-shop rather than going out to third-party vendors. Note for defense and aerospace buyers: B&R is not ITAR- or AS9100-registered, so export-controlled prints and AS9100 flow-down programs need a registered shop — we'll tell you that on the first call, not after the PO.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

Frequently asked questions

8 questions covering the Hwacheon Mega 100 platform and Industrial & General Manufacturing work. Also indexed as FAQ schema for AI answer engines.

What's special about the Mega 100's 10" thru-hole?

It accommodates long bar stock, tubing, and hollow parts that most CNC lathes can't clear. Downhole tool bodies with internal passages, coiled-tubing tool bodies, and long valve stems with bore access from both ends — parts most job shops outsource because their machines can't take the stock.

How long a part can the Mega 100 turn?

B&R's stated envelope is 100" (10 ft) between centers. The Mega-100 platform is available up to 6,000 mm (~20 ft) between centers on longer configurations.

What's the chuck swing and max cutting diameter?

Chuck swing over bed 42.5" (1,080 mm); max cutting Ø on the platform 27.5" (700 mm). Two-range spindle at 400/750 RPM with a 25 HP main motor.

What kind of industrial customers does B&R work with?

Industrial equipment OEMs, custom machinery builders, non-oilfield pump and valve manufacturers, food and pharma equipment suppliers, mining and heavy-equipment operators, and any customer who needs a legacy or reverse-engineered part made right.

Do you machine one-off replacement parts?

Yes — a lot of our industrial work is exactly this. Send the sample or drawing; if the material is on the shelf, we're often faster than the OEM. Reverse-engineered legacy parts across almost every sector.

What materials do you run for industrial work?

Full range — stainless (304, 316, 17-4 PH), tool steels, 4140/4340, brass, bronze, aluminum, cast iron, and specialty alloys per customer requirement. If you need something unusual, ask.

Do you offer emergency turnaround on down-machine parts?

Yes for repeat customers with material on the shelf. Same-day and next-day realistic when we already have the alloy. Call (936) 291-7827 and describe what's down.

Can you do small-batch production or is B&R just one-offs?

Both. Prototype and small-batch through repeat production. Small production runs where a big shop's minimum order would kill the economics are exactly where we fit.

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Ready to talk about your Industrial & General Manufacturing work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."

Published by B&R Productions — a precision CNC machining shop in New Waverly, Texas, in business since 1994. Serving oil & gas, aerospace, defense, and industrial customers across Texas and the Gulf Coast.

Written by the B&R Productions team. Machine specs verified against manufacturer data. Alloy and process notes reflect shop practice as of 2026-01-15.

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